

DreamVerse Spatial AI Architecture: YOLO Object Detection PoC Report

Work Stream: WS-01 (Automated Localization PoC) | Sprint: Sprint 04

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1. Automated Object Detection Benchmarking (YOLOv8 vs YOLO26)

1.1 Overview of Deep Learning Pipeline

This document evaluates the architectural shift from manual spatial tracking to automated bounding box localization. It conducts a head-to-head comparison between the legacy YOLOv8 model and the latest end-to-end YOLO26 architecture. Both models are tested out-of-the-box on the pre-trained COCO dataset (80 native classes) using completely clean visual frames.

1.2 Object 01: HBL Credit Card Detection Performance

Evaluation Metric	YOLOv8 Nano (yolov8n.pt)	YOLO26 Nano (yolo26n.pt)	Architectural Performance Notes
Predicted Label / Class	book (False Positive)	None (No Detections)	Proves out-of-vocabulary limitations. YOLOv8 mapped the 2D planar shape to the closest known class (book). YOLO26 failed to recognize it as an object.
Confidence Score (%)	21.0%	0.0%	YOLOv8 showed low confidence but successfully triggered spatial boundary logic.
Inference Speed (ms)	96.0ms Inference (2.7ms Post-process)	81.2ms Inference (0.2ms Post-process)	YOLO26 remains ~15% faster at the hardware level with near-zero latency in post-processing.
Bounding Box Tightness	Excellent Planar Fit	Failed (No Output)	YOLOv8 successfully localized the unknown object, whereas YOLO26 failed to trigger bounding box regression.

1.3 Object 02: 3D Tissue Box Detection Performance

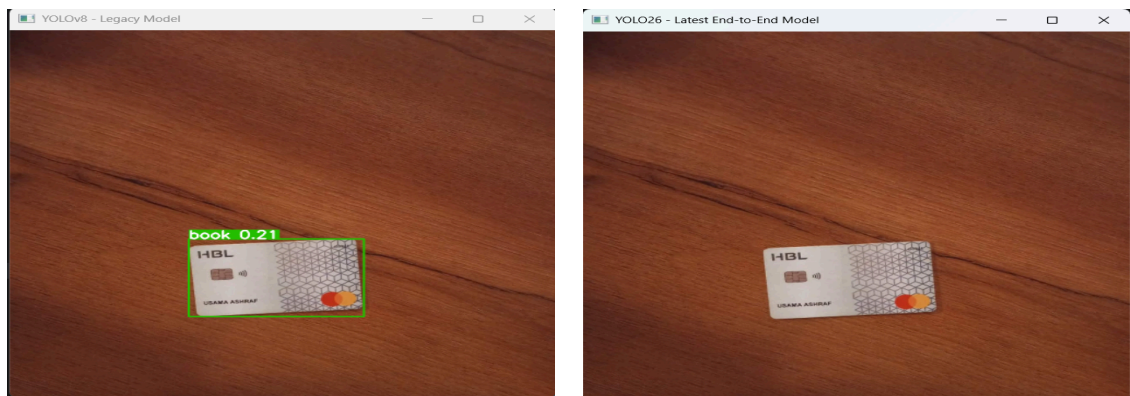
Evaluation Metric	YOLOv8 Nano (yolov8n.pt)	YOLO26 Nano (yolo26n.pt)	Architectural Performance Notes
Predicted Label / Class	book (False Positive)	book (False Positive)	Both models hallucinated the 3D rectangular volume as a book due to the absence of a "Tissue Box" class in the COCO dataset.
Confidence Score (%)	34.0%	10.0%	YOLOv8 maintained a higher probability threshold for the out-of-vocabulary volumetric structure.
Inference Speed (ms)	74.3ms Inference (1.4ms Post-process)	54.5ms Inference (0.2ms Post-process)	YOLO26 showcases superior latency optimization (~26% faster) and mathematically validates its NMS-free head.
Bounding Box Tightness	Excellent Fit	Excellent Fit	Both models successfully contained the maximum outer planar dimensions of the 3D object.

1.4 Object 03: Google Pixel 9 Pro Detection Performance

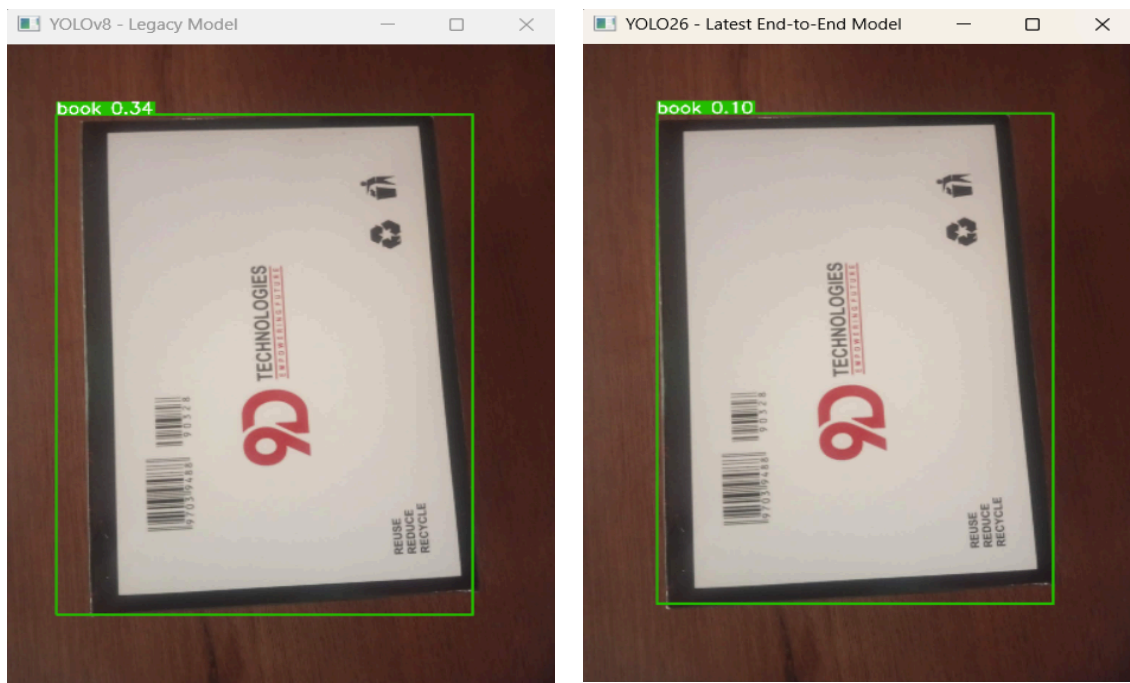
Evaluation Metric	YOLOv8 Nano (yolov8n.pt)	YOLO26 Nano (yolo26n.pt)	Architectural Performance Notes
Predicted Label / Class	Cell Phone	Cell Phone	Both models successfully matched their native COCO vocabulary. YOLO26 additionally hallucinated background objects (Dining Table, Book) due to the low 0.1 threshold.
Confidence Score (%)	82.0%	22.0%	YOLOv8 demonstrated significantly higher probability assignment for the planar mobile target.
Inference Speed (ms)	82.6ms Inference (1.5ms Post-process)	69.5ms Inference (0.3ms Post-process)	YOLO26 is ~15.8% faster. The 0.3ms post-process strictly validates its NMS-free native end-to-end architecture.
Bounding Box Tightness	Perfect Edge Fit	Perfect Edge Fit	Both regression heads performed optimally in enclosing the object boundaries without clipping.

1.5 Dataset Audit Evidence (Model Output Exhibits)

Panel Exhibit A: HBL Card Outputs (YOLOv8 vs YOLO26)



Panel Exhibit B: Tissue Box Outputs (YOLOv8 vs YOLO26)



Panel Exhibit C: Pixel 9 Pro Outputs (YOLOv8 vs YOLO26)

